

CLAUDE CODE · HEADLESS

Put an AI agent in your CI

Claude Code isn't just a chat window. It runs headless — in scripts, cron, and CI. Here's the playbook.

WHY

The shift

Stop copy-pasting into a chat box.
A headless agent runs on a
trigger, does the work unattended,
and reports back.

01 · THE CORE

Print mode

One flag turns Claude Code into a scriptable command. Pipe anything in.

- `claude -p 'fix the failing test'`
- `cat error.log | claude -p 'explain this'`

02 · PIPE IT ANYWHERE

Structured output

Get JSON back — including the run cost — so scripts can act on the result.

- `--output-format json`
- `jq '.total_cost_usd' → 0.0024`

03 · SAFETY FIRST

Scope what it can do

Never give a headless agent a blank check. Whitelist exact tools and cap the run.

- `--allowedTools 'Bash(npm test),Read,Edit'`
- `--permission-mode acceptEdits`
- `--max-turns 3`

04 · THE AGENT SDK

Or build it in

Embed the agent in your own app
— TypeScript or Python — in one
call.

- `npm i @anthropic-ai/claude-agent-sdk`
- `pip install claude-agent-sdk`
- `query({ prompt, options })`

05 · CI IN ~6 LINES

Drop it in GitHub Actions

The official action + your API key
= an agent on every pull request.

- uses: anthropics/claude-code-action@v1
- secret: ANTHROPIC_API_KEY

USE CASES

Real jobs to automate

What teams ship headless today:

- PR review on every push
- Issue triage + auto-labels
- Nightly changelog from git log
- Issue → PR implementation

[SAVE THIS](#)

The cheat sheet

The safe headless one-liner:

- `claude -p 'task' --output-format json`
- `--allowedTools 'Bash(npm test),Read' --max-turns 3`



DO THIS TODAY

Where teams get stuck

The prompt is easy. The safe, reliable harness around it is the hard part — that's what we build at Noblerr. Start small: wrap one repetitive task as `claude -p` today.

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